Simon Memory Game Using ESP32

Project Description

The **Simon Memory Game** is an interactive embedded systems project developed using the **ESP32 microcontroller** to evaluate and improve a player's short-term memory and reaction skills. The system generates a random sequence of illuminated LEDs accompanied by audio signals from a buzzer. The player must accurately reproduce the displayed sequence by pressing the corresponding push buttons.

The game begins with a simple one-step sequence and progressively increases in difficulty by adding an additional step after each successful round. If the player enters an incorrect sequence, the game provides visual and audio feedback, displays the final score or level achieved (if an OLED display is used), and automatically resets for a new game.

This project demonstrates key concepts in embedded systems, including digital input/output control, random sequence generation, user interaction, timing functions, audio-visual feedback, and game logic implementation. It is an excellent educational project for learning ESP32 programming, electronics, and real-time embedded application development.

Hardware Components

Component	Quantity	Purpose
ESP32 Development Board	1	Main microcontroller responsible for game logic and control
LEDs	4	Display the memory sequence visually
Push Buttons	4	User input for repeating the sequence
Passive Buzzer	1	Provides sound effects and feedback
220Ω Resistors	4	Current limiting for LEDs
10kΩ Resistors	4	Pull-up or pull-down resistors for push buttons
OLED Display (Optional)	1	Displays score, level, and game status
Breadboard	1	Circuit prototyping
Jumper Wires	As Required	Electrical connections
USB Cable (CH340 Compatible)	1	Programming and power supply for the ESP32
5V / 3.3V Power Supply	1	Powers the complete system

Key Features

- Random memory sequence generation
- Progressive difficulty with increasing sequence length
- Interactive LED and buzzer feedback
- Push-button user interface
- Automatic score and level progression
- Game-over detection and reset functionality
- Optional OLED display for real-time score and status
- Built using the ESP32 platform with efficient embedded programming

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Applications

- Memory and concentration training
- Embedded systems learning
- Microcontroller programming practice
- Educational electronics projects
- STEM demonstrations
- Interactive gaming systems